

2.3 Semiconductor Models

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2.3.3 Semiconductor Band Model

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Literature: Pierret, Chapter 2.1-2.3, page 23-40
Pierret, Appendix A, page 733-748

2.3.1 Electron States in Atoms

– Formalism of Quantum Mechanics –

- Systems with **atomic dimensions**, such as the electrons in a semiconductor atom, are described by the **quantum mechanics** and not the classical “Newtonian” mechanics
- The quantum mechanics leads to the concept of quantized energy values for the electrons of an atom, necessary to explain e.g. the discrete spectral lines emitted by heated gases
- The (time-dependent) Schrödinger equation describes the dynamic behavior of a single-particle system, e.g. the behavior of an electron in the potential of the hydrogen H^+ nucleus

$$-\frac{\hbar^2}{2m} \nabla^2 \Psi + U(x, y, z) \Psi = -\frac{\hbar}{i} \frac{\partial \Psi}{\partial t}$$

m is the particle mass, U the system's potential energy and $i = \sqrt{-1}$

- The complex wavefunction $\Psi = \Psi(x, y, z, t)$ describes the dynamic behavior of the particle in the potential U

Formalism of Quantum Mechanics (cont.)

- The product $\Psi^* \Psi dV$ of the wavefunction Ψ and its complex conjugate Ψ^* gives the probability to find the particle in a certain volume dV

$$\iiint_V \Psi^* \Psi dV = 1$$

- The expectation values for the particles position and momentum can be calculated from

$$\langle x \rangle = \iiint \Psi^* x \Psi dV$$

$$\langle p_x \rangle = \iiint \Psi^* \frac{\hbar}{i} \frac{\partial \Psi}{\partial x} dV$$

- Assuming a single-particle system with fixed total energy E , we can simplify the Schrödinger equation by separating the wave function as follows

$$\Psi(x, y, z, t) = \psi(x, y, z) e^{-iEt/\hbar}$$

yielding

$$\frac{\hbar^2}{2m} \nabla^2 \psi + [E - U(x, y, z)] \psi = 0$$

The Hydrogen Atom

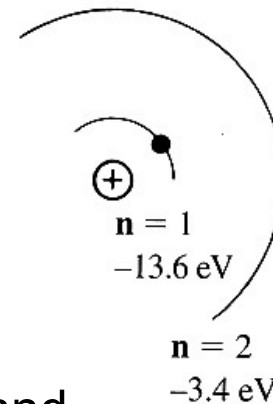
- Solving the Schrödinger equation for an electron in the electrostatic potential U of the hydrogen nucleus

$$U = -\frac{q^2}{4\pi\epsilon_0 r}$$

yields the wavefunctions Ψ_{nlm} of the different electron states of the hydrogen atom and their energies

$$E_n = -\frac{13.6}{n^2} \text{ eV}$$

- The electron state is described by three quantum numbers n, l and m (and spin s)
- Even mathematically simpler models, such as an electron in a 1-D potential barrier, result in a quantized energy



$n=3$
 $E_H = -1.51 \text{ eV}$

Pierret, Fig. 2.1

Note: $1 \text{ eV} = 1.6 \cdot 10^{-19} \text{ Joules}$

Multi-Electron Atoms

- Similar to the hydrogen atom, the electron states in multi-electron atoms are uniquely characterized by **4 quantum numbers n , l , m and s**

$n = 1, 2, 3, \dots$	Principal quantum number
$l = 0, 1, 2, \dots, n-1$ $l = s, p, d, f, \dots$	Azimuthal quantum number
$m = -l, -l+1, \dots, l$	Magnetic orbital quantum number
$s = -1/2 \text{ and } 1/2$	Spin quantum number

n	1	1	2	2	2	2	2	2	2	2	3	3	3	3	3	3	3
l	0	0	0	0	1	1	1	1	1	1	0	0	1	1	1	1	1
m	0	0	0	0	-1	-1	0	0	1	1	0	0	-1	-1	0	0	1
s	$\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$
State	1s	1s	2s	2s	2p	2p	2p	2p	2p	2p	3s	3s	3p	3p	3p	3p	3p

Pierret, Tab. A.3

- Pauli Exclusion Principle:** No two electrons in a system can be characterized by the same set of quantum numbers
- Thus, at $T = 0K$, the electron states are filled from the one with the lowest energy on with the available electrons

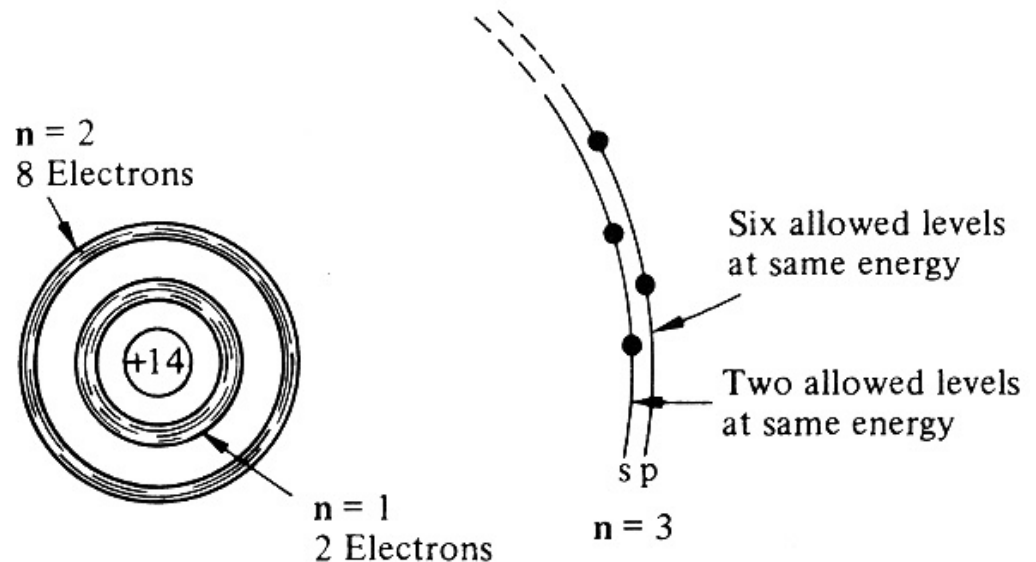
Energy States and Electronic Configuration in Elements 1-14

Atomic Number	Element	Filled States														Electronic Configuration
		1s	1s	2s	2s	2p	2p	2p	2p	2p	2p	3s	3s	3p	3p	
1	H	↑														1s
2	He	↑	↓													1s ²
3	Li	↑	↓	↑												1s ² 2s
4	Be	↑	↓	↑	↓											1s ² 2s ²
5	B	↑	↓	↑	↓	↑										1s ² 2s ² 2p
6	C	↑	↓	↑	↓	↑	↓									1s ² 2s ² 2p ²
7	N	↑	↓	↑	↓	↑	↓	↑								1s ² 2s ² 2p ³
8	O	↑	↓	↑	↓	↑	↓	↑	↓							1s ² 2s ² 2p ⁴
9	F	↑	↓	↑	↓	↑	↓	↑	↓	↑						1s ² 2s ² 2p ⁵
10	Ne	↑	↓	↑	↓	↑	↓	↑	↓	↑	↓					1s ² 2s ² 2p ⁶
11	Na	↑	↓	↑	↓	↑	↓	↑	↓	↑	↓	↑				1s ² 2s ² 2p ⁶ 3s
12	Mg	↑	↓	↑	↓	↑	↓	↑	↓	↑	↓	↑	↓			1s ² 2s ² 2p ⁶ 3s ²
13	Al	↑	↓	↑	↓	↑	↓	↑	↓	↑	↓	↑	↓	↑		1s ² 2s ² 2p ⁶ 3s ² 3p
14	Si	↑	↓	↑	↓	↑	↓	↑	↓	↑	↓	↑	↓	↑	↓	1s ² 2s ² 2p ⁶ 3s ² 3p ²

Pierret, Tab. A.3

Silicon Atomic Structure

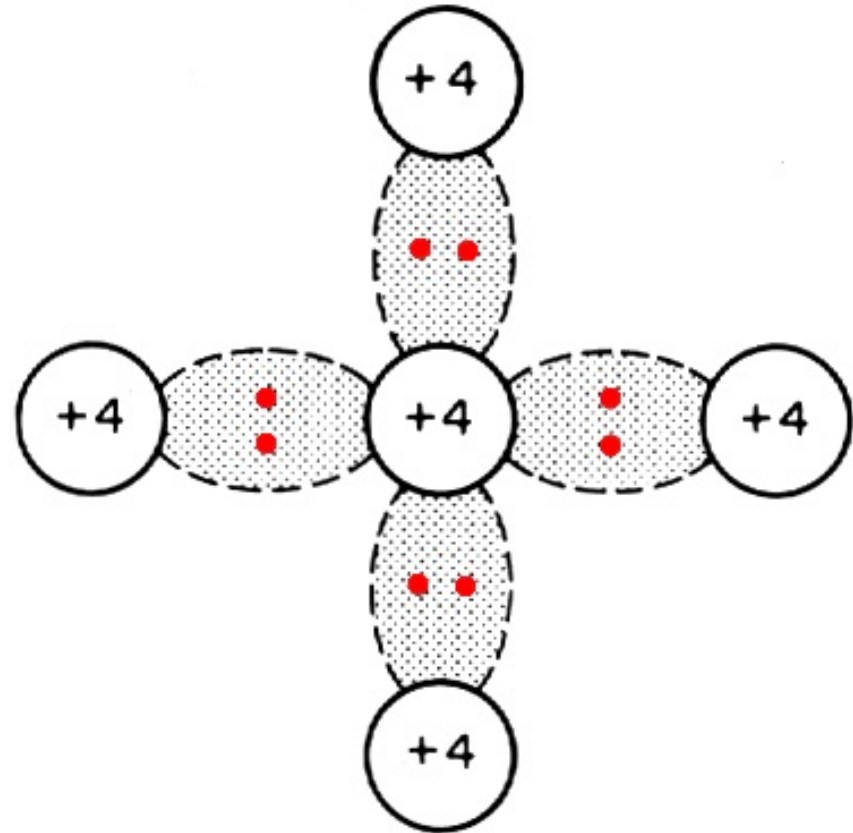
- Silicon with its 14 electrons has (in its ground state) filled $n = 1$ (2 electrons) and $n = 2$ (8 electrons) levels; these are deep-lying energy levels, tightly bound to the nucleus
- In case of the $n = 3$, the possible states (2 s- and 6 p-states) are filled with the 4 remaining electrons; these 4 rather weakly bound electrons are called **valence electrons** and participate in chemical reactions and atom-atom interaction



Pierret, Fig. 2.2

2.3.2 Semiconductor Bond Model

- Bonding to next neighbors is achieved by sharing valence electrons (electrons in outermost shell), forming covalent bonds
- In case of silicon:
 - Atomic number = number of electrons = 14
 - Number of valence electrons = 4
 - Covalent tetraeder bonding (diamond lattice)

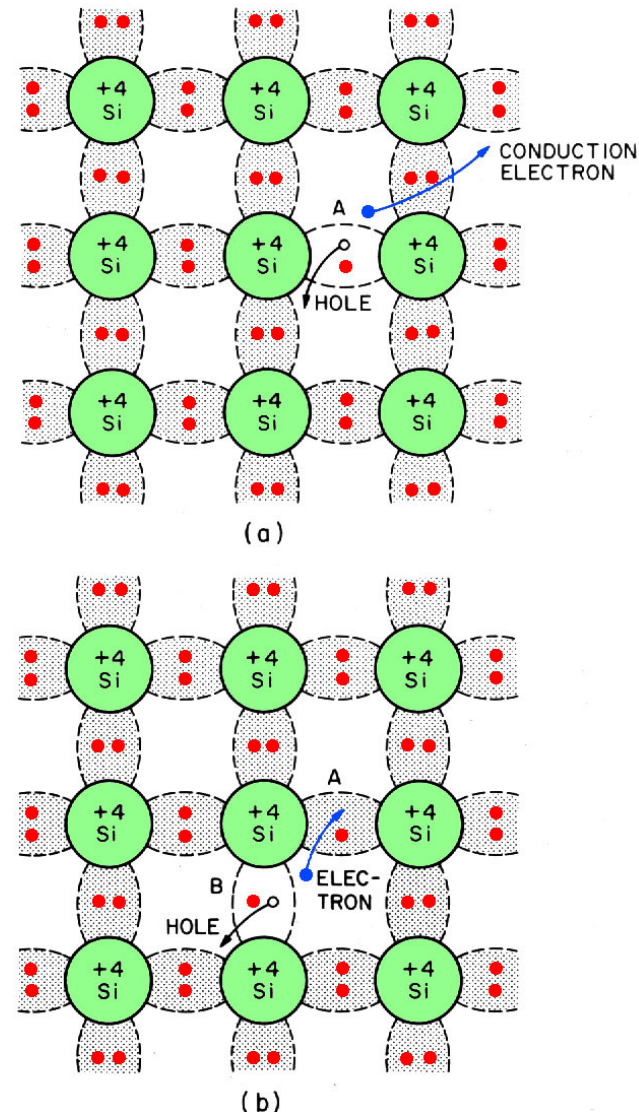


Sze, Fig. 2.11

Semiconductor Bond Model

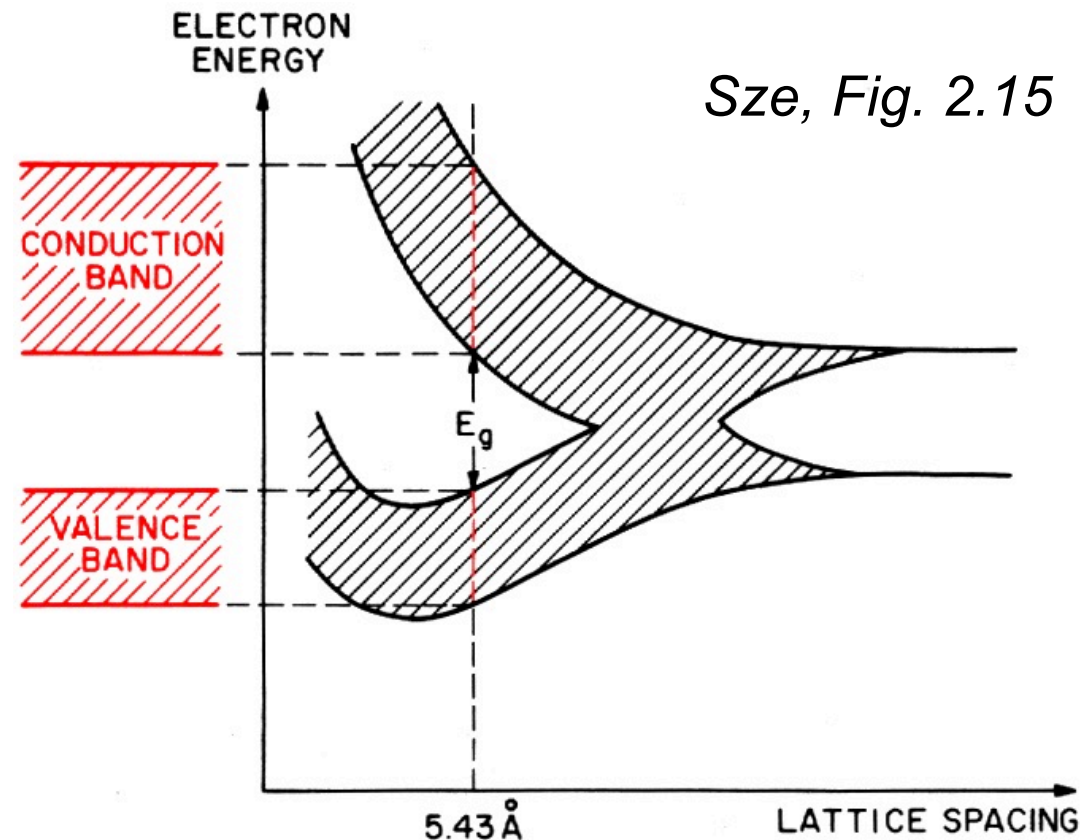
- Valence electrons are **weakly bound** (compared to electrons in inner shells)
- Thermal energy at room temperature can break covalent bonds and “free” electrons which now contribute to the materials conduction
- Remaining are “missing” bonds/electrons, which are called holes (and also contribute to the conduction, **but why?**)

Sze, Fig. 2.12

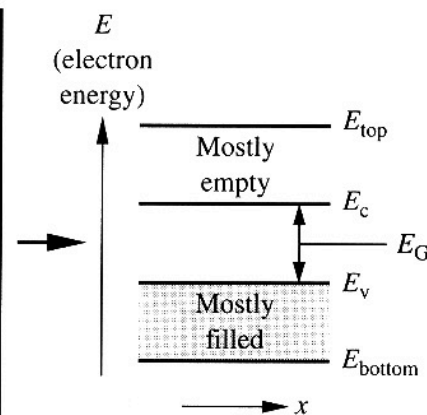
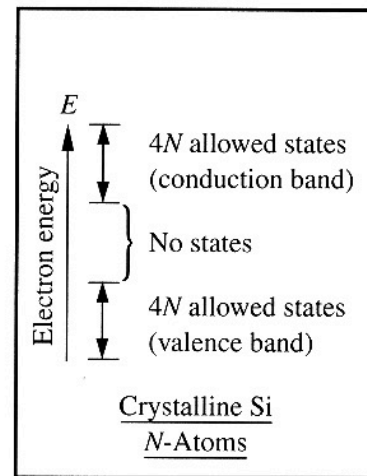
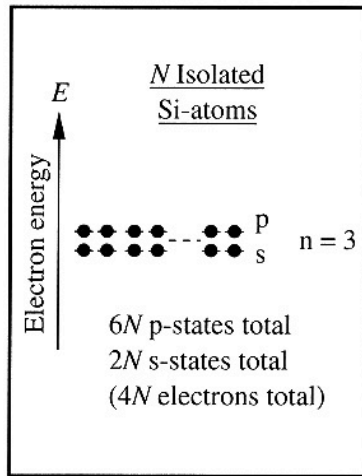


2.3.3 Semiconductor Band Model

- **Single atom:** according to Bohr's atom model, the electrons occupy well defined, **discrete energy levels**

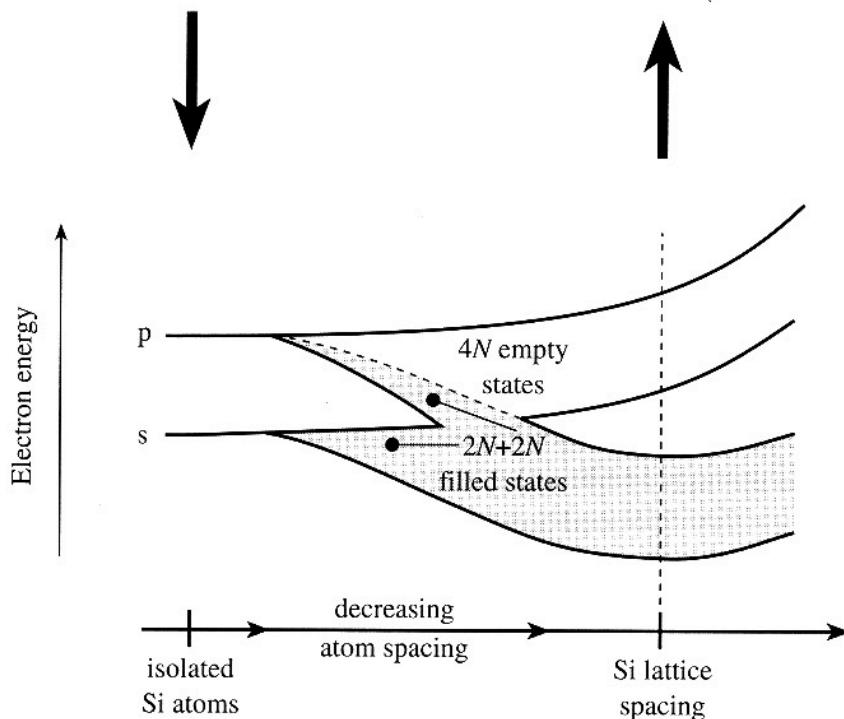


- **Crystal:** due to the interaction of neighboring atoms, the discrete energy levels split up and a **bandstructure** is developing featuring **allowed and forbidden energy bands**



From Isolated Atoms to a Crystal

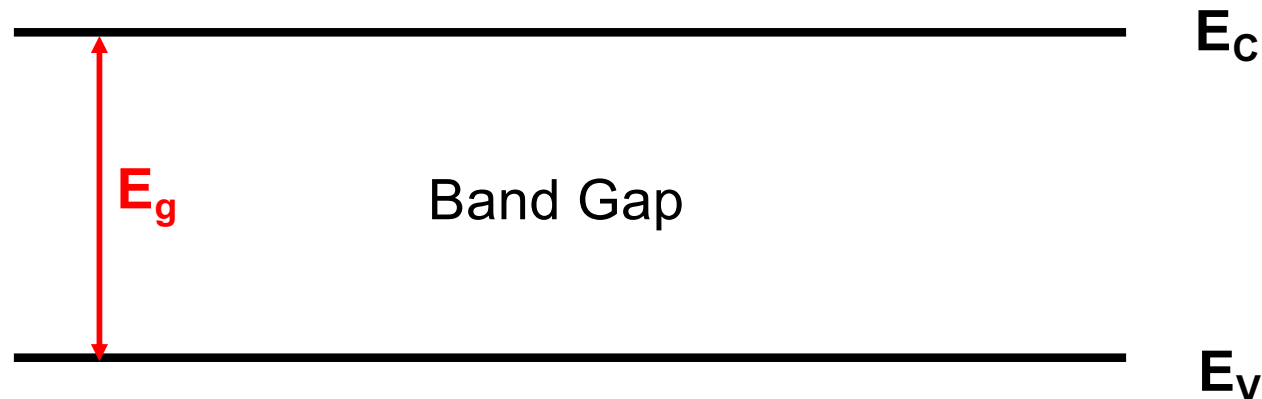
- Semiconductor crystal has **valence band** and **conduction band** separated by **bandgap E_g**
- At $T = 0$ K:
 - Completely filled valence band
 - Completely empty conduction band
- At elevated T , some electrons can be excited from the valence to the conduction band
- Note: Electrons in bands are not associated with particular atom



Pierret, Fig. 2.5

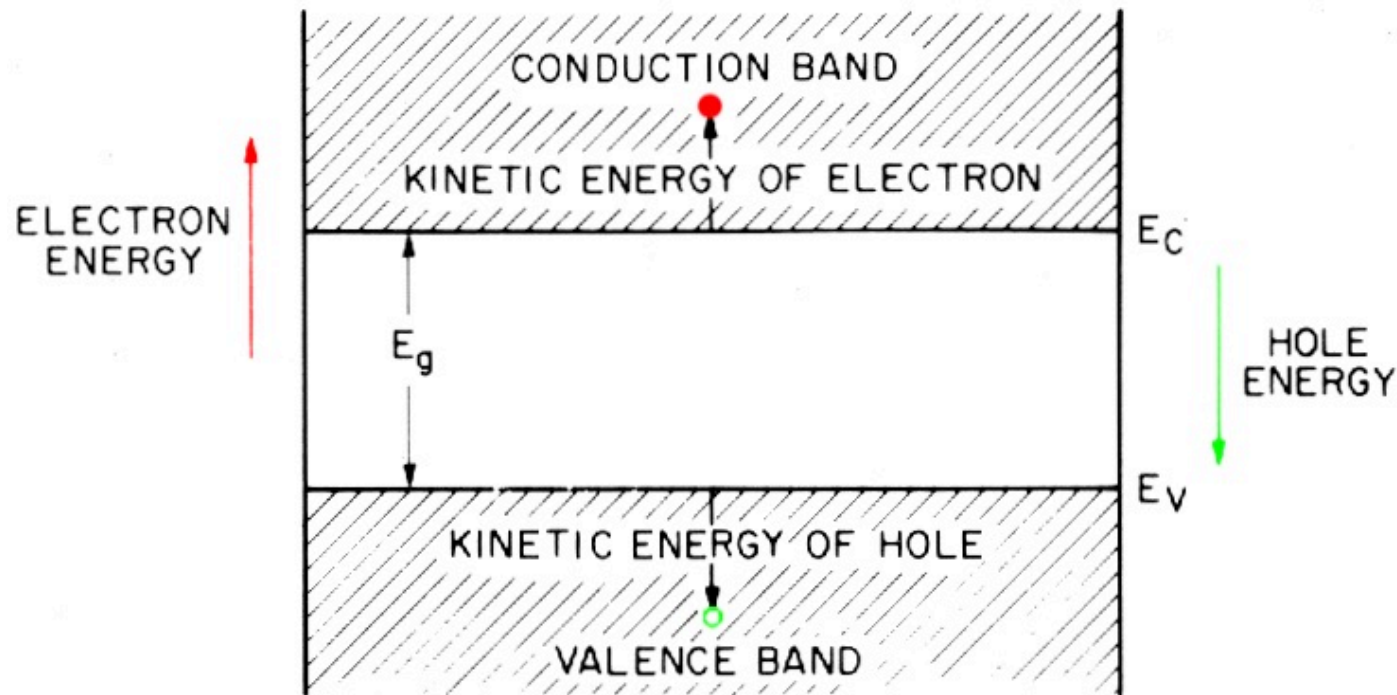
Semiconductor Band Model

- Reason for energy level splitting is the **Pauli principle** stating that no 2 electrons can occupy the same state (with respect to energy, momentum, spin), i.e. have the same quantum numbers n , l , m , and s
- Simplified Model: **valence and conduction band are separated by a band gap**; E_C is the lowest energy of the conduction band, E_V the highest energy of the valence band

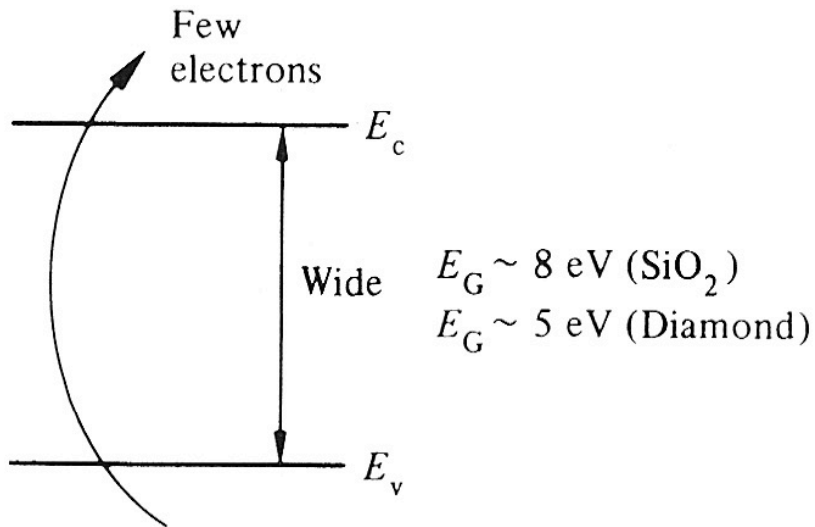


Carriers: Electrons and Holes

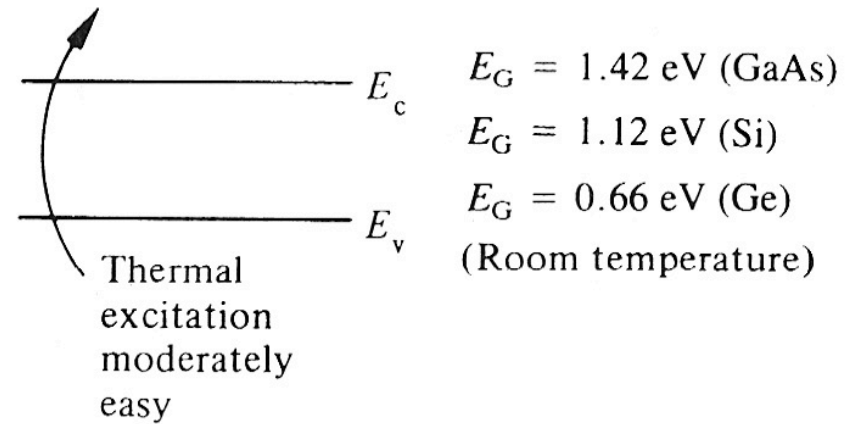
- Completely filled band and empty band do not contribute to current conduction
- If an **electron is excited** from the valence into the conduction band, the additional **electron** in the conduction band contributes to the current conduction; similarly, the missing electron in the valence band (like a bubble in a liquid), a so-called **hole**, contributes to the current conduction; the (thermal) excitation has created an **electron/hole pair**



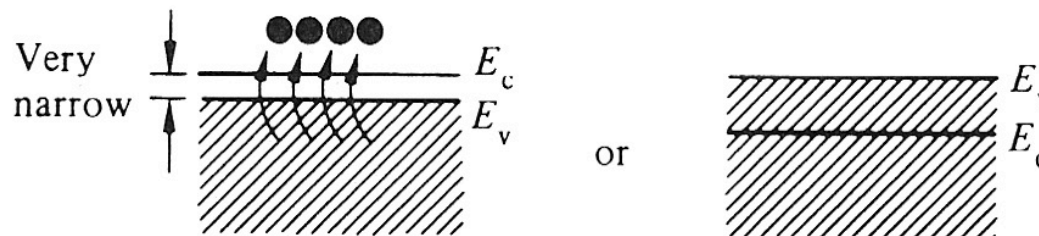
Band Gap & Material Classification



Insulator



Semiconductor



Metal

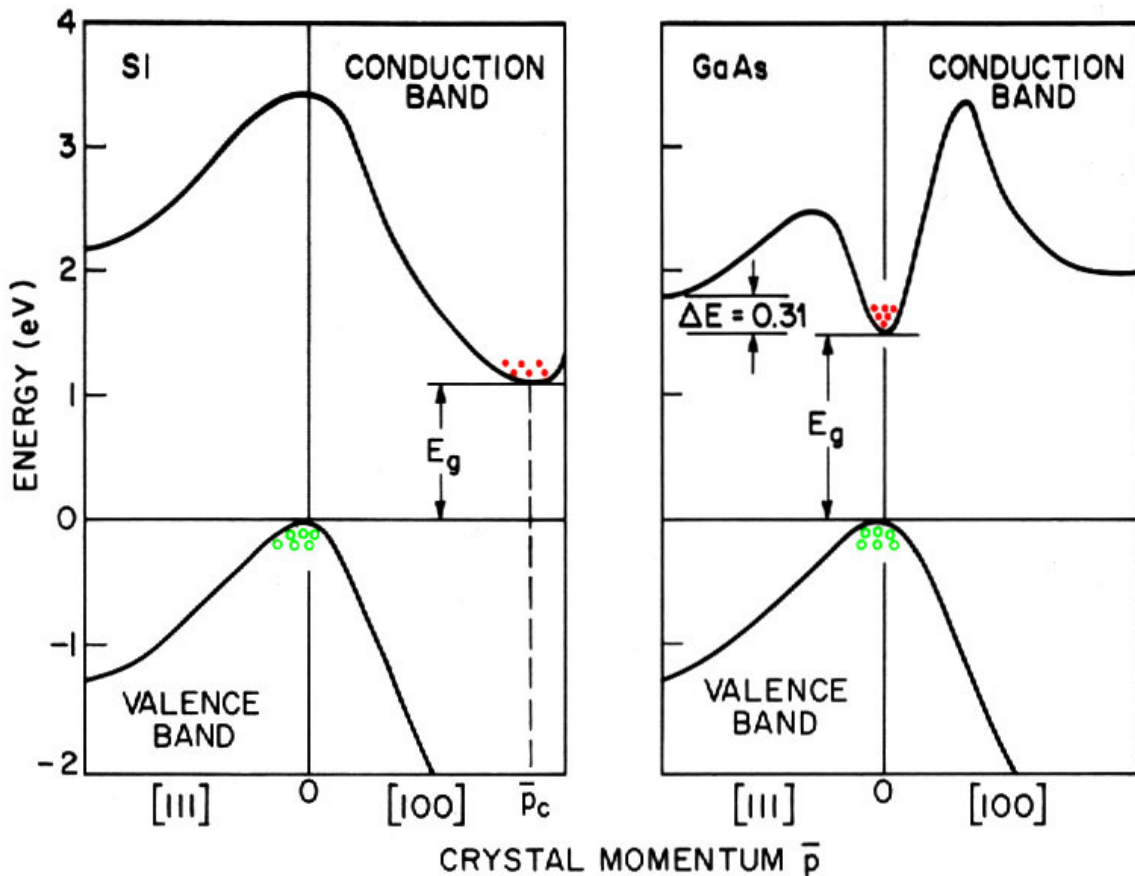
Pierret, Fig. 2.8

“Real” Derivation of Band Structure

- Using **quantum mechanics** by solving the **3-D Schrödinger equation** in the **periodic potential** of the nuclei
- Numerical calculations deliver the band diagrams, i.e. the $E(k)$, $E(p)$ relationships with **wavenumber k** and **crystal momentum p**

$$\mathbf{p} = \hbar \mathbf{k}$$

Si and GaAs Simplified Bandstructure

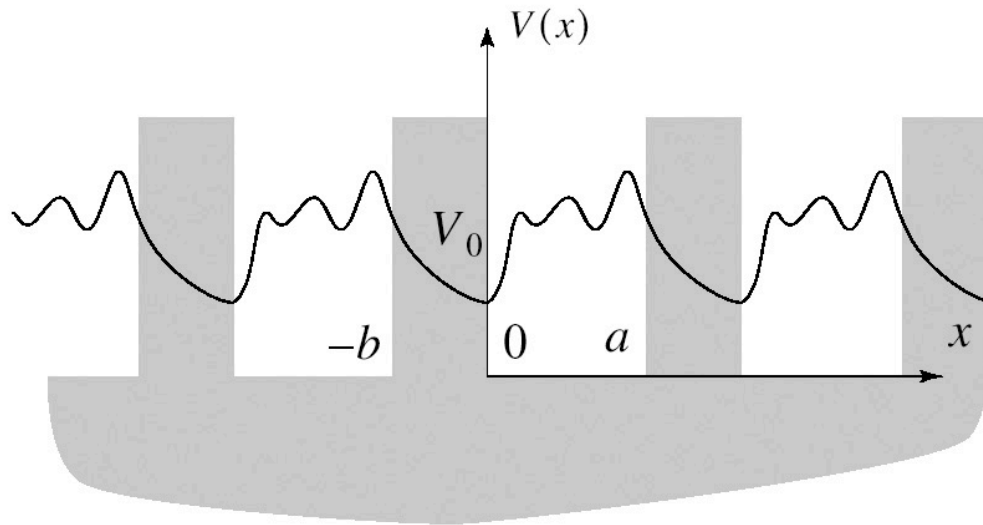


Sze, Fig. 2.18

Origin of Band Structure

– Kronig-Penney Model –

- Electron in a **periodic potential** results in a band structure with allowed and forbidden energy bands
- The Kronig-Penney model is a **1-D analytical model** for an electron in a periodic potential, resulting in a band structure
- Periodic potential of Kronig-Penney model:

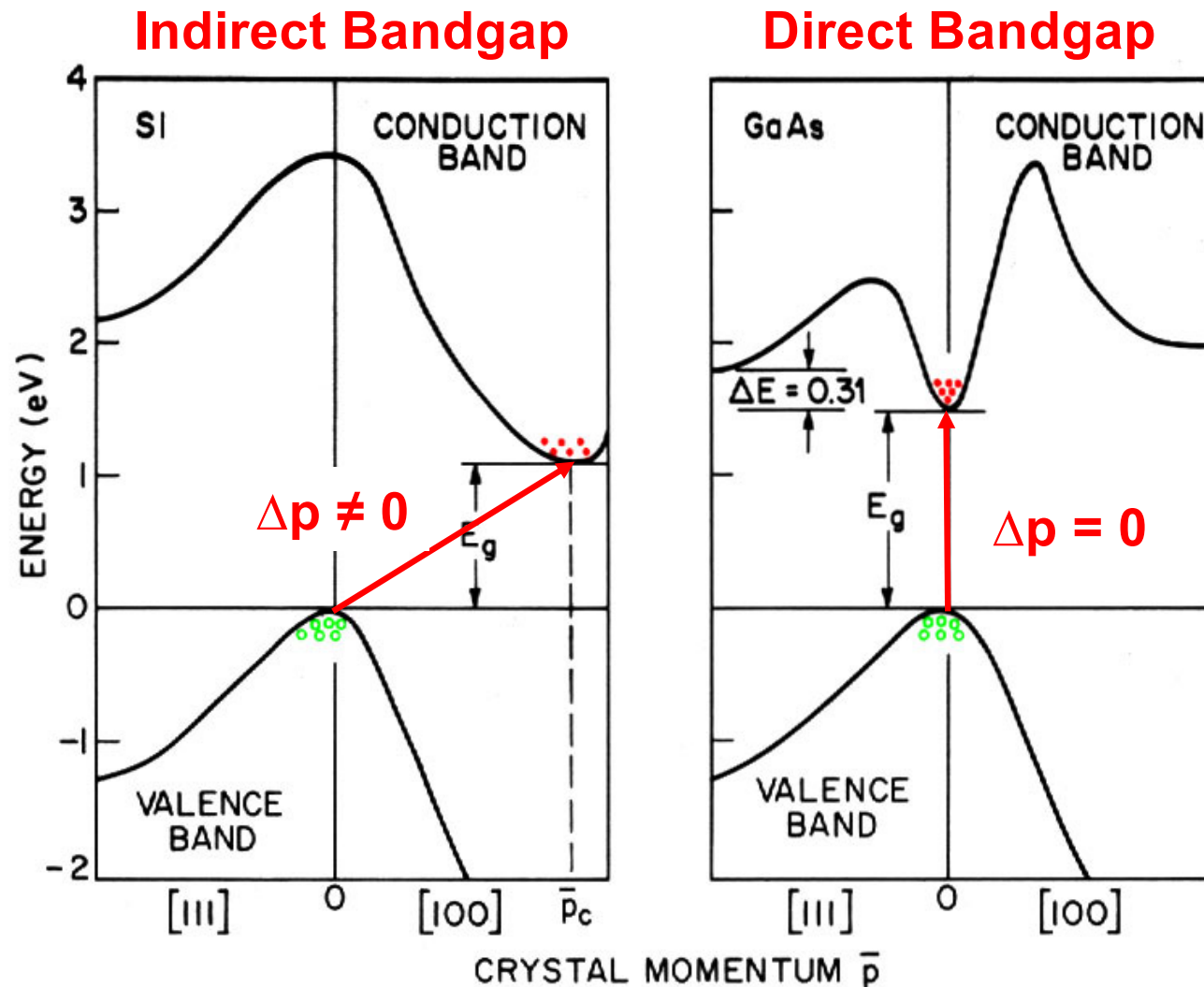


Korvink, Fig. 3.10

- For details, see e.g.: Korvink, Greiner, Semiconductors for Micro- and Nanotechnology, Wiley-VCH, 2003, Chapter 3.3

Indirect vs. Direct Bandgap Semiconductor

– Simplified Si & GaAs Band Structure –



Sze, Fig. 2.18

Characteristics of Si and GaAs

Silicon Si

- Element semiconductor
- Diamond lattice structure
- Indirect Semiconductor
- Generation of electron-hole pairs requires ΔE and Δp
- Bandgap: $E_g = 1.12$ eV at room temperature
- Lattice constant: $a = 5.43$ Å

Gallium Arsenide GaAs

- III-V semiconductor
- Zincblende lattice structure
- Direct Semiconductor
- Generation of electron-hole pairs requires ΔE only, making GaAs suitable for photonic applications
- Bandgap: $E_g = 1.42$ eV at room temperature
- Lattice constant: $a = 5.65$ Å

The figure is a plot of Energy [eV] versus Wave vector coordinate. The y-axis ranges from -12 to 6 eV with major ticks every 2 eV. The x-axis shows high-symmetry points: L, Λ , Γ , Δ , X, U, K, Σ , and Γ . Vertical gray lines connect these points to the bands. Labels for bands include L_3 , L_1 , L_3' , L_1 , L_2' , Γ_2' , Γ_{15} , Γ_{25}' , X_1 , X_4 , Γ_1 , and Γ_{25}' . Two horizontal red lines are drawn at approximately 1.2 eV and 0 eV.

Simplification of $E(k)$ Band Structure within ECE 3040

- Essential part of band structure:
 - Top of valence band with energy E_v
 - Bottom of conduction band with energy E_c
- Approximation of band structure around the band extrema by parabolic $E(k)$ relationship

$$E(k) = \frac{\hbar^2 k^2}{2m_{\text{eff}}}$$

with the effective mass m_{eff}

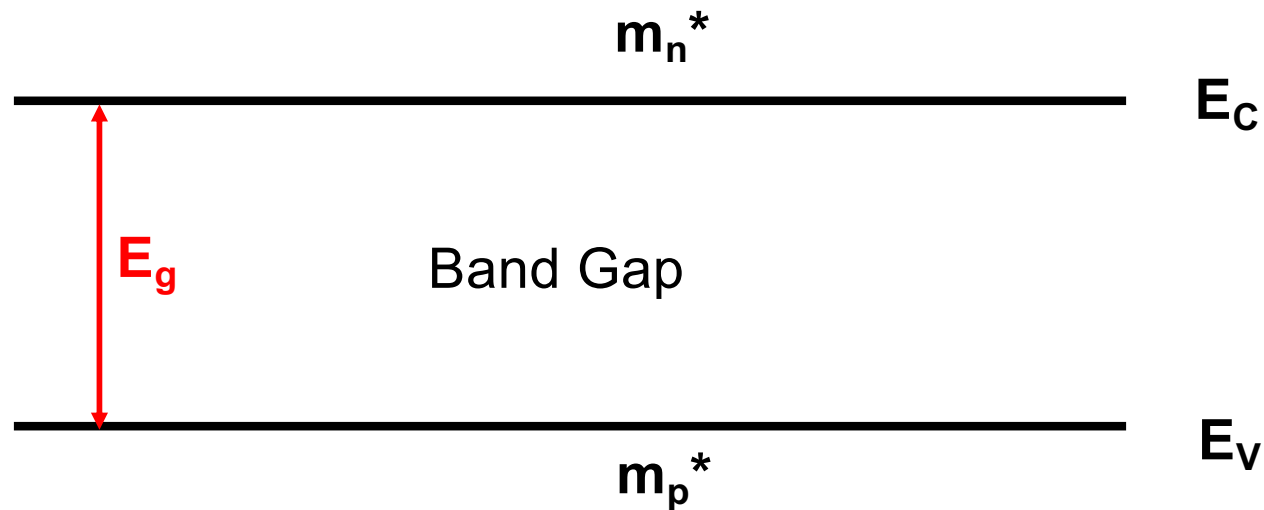
Note: free electron

$$\left. \begin{array}{l} p = m_0 v \\ E_{\text{kin}} = \frac{1}{2} m_0 v^2 \\ p = \hbar k = h/\lambda \end{array} \right\} E = \frac{\hbar^2 k^2}{2m_0}$$

- Electron behavior in the crystal is similar to free electron behavior with

$$m_e = m_{\text{eff}} \neq m_0$$

Simplified Semiconductor Bandstructure



Effective Mass m_{eff} (m_n^* , m_p^*)

- Definition 1-D:

$$m_{\text{eff}}^{-1} = \frac{1}{\hbar^2} \frac{\partial^2 E(k)}{\partial k^2}$$

- Definition 3-D:

$$\left(\frac{1}{m_{\text{eff}}} \right)_{ij} = \frac{1}{\hbar^2} \frac{\partial^2 E(k)}{\partial k_i \partial k_j}$$

- Electron kinetics in single-crystal material is similar to free electron behavior but with effective mass m_{eff}

Effective Mass in Si, Ge & GaAs

Material	m_n^*/m_0	m_p^*/m_0
Si	1.18	0.81
Ge	0.55	0.36
GaAs	0.066	0.52

- Free electron mass: $m_0 = 9.1 \cdot 10^{-31}$ kg
- Shown are average electron and hole effective masses as **needed for the density of states** at 300 K (see Chapter 2.5), but keep in mind that m_n^* and m_p^* are tensors!

Effective Mass Concept

- The movement of carriers (electrons and holes) inside the semiconductor crystal can be described by classical (Newtonian) mechanics relations just by replacing the free-electron-mass m_0 with the effective mass m_n^* (or m_p^* for holes)

$$\mathbf{F} = m_n^* \frac{d\mathbf{v}_n}{dt} \quad \text{and} \quad \mathbf{F} = m_p^* \frac{d\mathbf{v}_p}{dt}$$

- The effective masses are however tensors, i.e. the carrier acceleration generally varies with direction of travel in the crystal

2.4 Semiconductor Doping

Example: **Silicon Doping:**

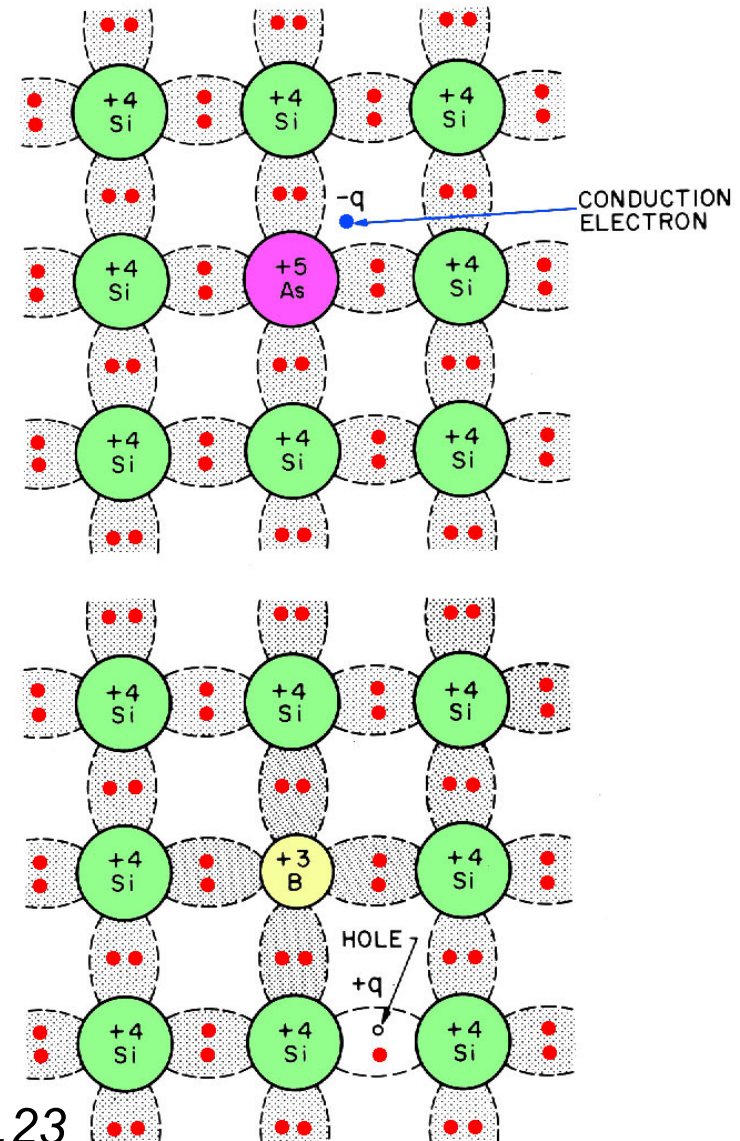
- Doping with **donor** atoms, **phosphorous P** or **arsenic As**, having both 5 valence electrons, yielding **N-type silicon**
- Doping with **acceptor** atoms, **boron B**, having 3 valence electrons, yielding **P-type silicon**
- **With doping, the electrical conductivity of the semiconductor material is adjusted over a wide range**

boron 5 B 10.811(7)	carbon 6 C 12.0107(8)	nitrogen 7 N 14.00674(7)
aluminium 13 Al 26.981538(2)	silicon 14 Si 28.0855(3)	phosphorus 15 P 30.973761(2)
gallium 31 Ga 69.723(1)	germanium 32 Ge 72.64(1)	arsenic 33 As 74.92160(2)
indium 49 In 114.818(3)	tin 50 Sn 118.710(7)	antimony 51 Sb 121.760(1)
thallium 81 Tl 204.3833(2)	lead 82 Pb 207.2(1)	bismuth 83 Bi 208.98038(2)

Semiconductor Doping

Doping in **Bond Model**:

- **N-Type Doping:**
fifths valence electron does not participate in covalent bonds to next neighbors; only small energy is required to “free” this electron, thus contributing to the materials conductivity
- **P-Type Doping:**
due to doping atom with three valence electrons, one electron is missing for the covalent bonds to the four next neighbors; this “missing electron” (called a hole) can hop from covalent bond to bond, thus also contributing to the materials conductivity



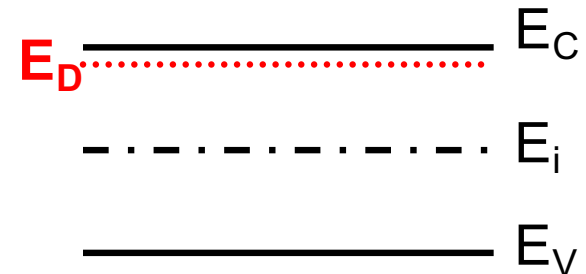
Sze, Fig. 2.23

Semiconductor Doping

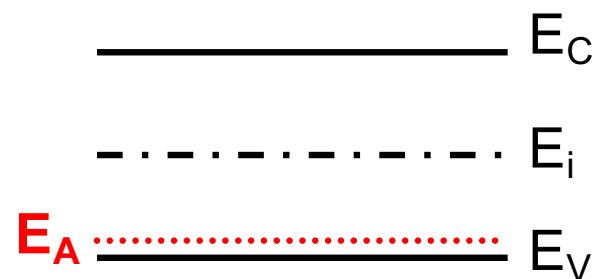
Doping in **Band Model**:

- Doping atoms distort the periodicity of the single-crystal crystal structure, thus **creating allowed (localized) energy states in the band gap**
- **N-Type Doping**: additional electron states E_D close to the bottom of the conduction band E_C ; at room temperature, the electrons from these states are excited into the conduction band, increasing the conductivity
- **P-Type Doping**: additional electron states E_A close to the top of the valence band E_V ; at room temperature, these states are occupied by electrons from the valence band, increasing the number of holes in the valence band and, thus, the conductivity

$E_C - E_D$ resp. $E_A - E_V$
 $\approx kT = 0.026 \text{ eV}$
 at room temperature



P in Si: $E_C - E_D = 0.045 \text{ eV}$



B in Si: $E_A - E_V = 0.045 \text{ eV}$

Carrier Concentration

Intrinsic Semiconductor (no doping: $N_A, N_D = 0$)

- Carrier Concentration: $n = p = n_i$

Extrinsic Semiconductor (doping concentration $N_A, N_D \neq 0$)

- Assumption: **Complete Ionization** (see Chapter 2.5)
- Majority Carrier Concentrations:
 - **N-Type Semiconductor:**
Density of donor atoms: N_D [cm^{-3}]
Density of electrons ($N_D \gg n_i$) : $n \approx N_D$ [cm^{-3}]
 - **P-Type Semiconductor:**
Density of acceptor atoms: N_A [cm^{-3}]
Density of holes ($N_A \gg n_i$) : $p \approx N_A$ [cm^{-3}]